Dataflow Analysis (2)

CSE552 Program Analysis — Lecture 8

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Live Variable Analysis

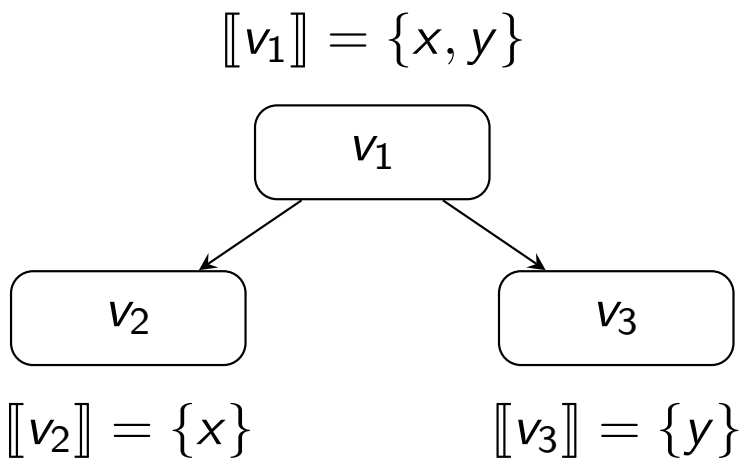
- A variable is *live* at a program point if there exists an execution where its value is read later in the execution without it being written to in between

```
// nothing is live
x = input();
// x is live
if input() {
  // x is live
  y = x;
  // y is live
} else {
  // nothing is live
  y = 1;
  // y is live
}
// y is live
z = y;
```

Live Variable Analysis — Motivation

- We can approximate the set of live variables using dataflow analysis
  - Application: register allocation
  - We want: the answer "not live" can be trusted and "live" is safe but useless

Live Variable Analysis — Abstract States



- State $= (\mathcal{P}(\text{Var}), \subseteq)$ — Power set lattice
- For each CFG node v, $[\![v]\!]$ denotes the set of variables live before the node
- $\text{JOIN}(v) = \bigsqcup_{u \in \text{succ}(v)} [\![u]\!] = \bigcup_{u \in \text{succ}(v)} [\![u]\!]$
  - This combines abstract states from the **successors**

Live Variable Analysis — Constraint Rule (Assignment)

- x=e: $[\![v]\!] = \text{JOIN}(v) \setminus \{x\} \cup \text{vars}(e)$

```
// y and z are live
x = y + z;
// x is live
```

Live Variable Analysis — Constraint Rules (Remaining)

- `if` x: $[\![v]\!] = \mathrm{JOIN}(v) \cup \{x\}$
- `entry`: $[\![v]\!] = \mathrm{JOIN}(v)$
- `return`: $[\![v]\!] = \mathrm{JOIN}(v) = \emptyset$

Available Expression Analysis

- A nontrivial expression (not a literal, not a variable) in a program is *available* at a program point if its current value has already been computed earlier in the execution

```
// nothing is available
x = y + 1;
// y + 1 is available
if input() {
  // y + 1 is available
  y = z + 1;
  // z + 1 is available
} else {
  // y + 1 is available
  x = z + 1;
  // y + 1 and z + 1 are available
}
// z + 1 is available
w = (z + 1) + (y + 1);
```

Available Expression Analysis — Motivation

- We can approximate the set of available expressions using dataflow analysis
  - Application: optimization (eliminating redundant computations)
  - We want: the answer "available" can be trusted and "not available" is safe but useless

Available Expression Analysis — Optimization Example

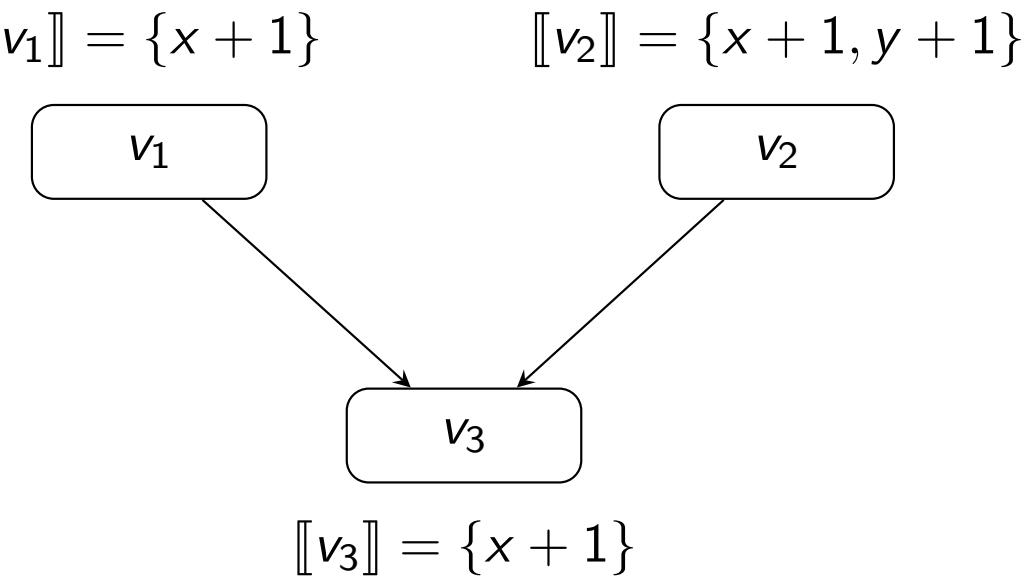
Before:

```
x = y + 1;
if input() {

  y = z + 1;
} else {

  x = z + 1;
}
w = (z + 1) + (y + 1);
```

After:

```
x = y + 1;
if input() {
  zplus1 = z + 1;
  y = zplus1;
} else {
  zplus1 = z + 1;
  x = zplus1;
}
w = zplus1 + (y + 1);
```

Available Expression Analysis — Abstract States



- State $= (\mathcal{P}(\text{Expr}), \supseteq)$ — **Reverse** power set lattice
- For each CFG node v, $[\![v]\!]$ denotes the set of expressions available after the node
- $\text{JOIN}(v) = \bigsqcup_{u \in \text{pred}(v)} [\![u]\!] = \bigcap_{u \in \text{pred}(v)} [\![u]\!]$
  - This combines abstract states from the **predecessors**

Available Expression Analysis — Constraint Rule (Assignment)

- x=e: $[\![v]\!] = (\text{JOIN}(v) \cup \text{exprs}(e))\downarrow x$
  - $\downarrow x$ removes all expressions containing x
  - exprs collects all nontrivial expressions

$$\text{exprs}(x) = \emptyset$$
$$\text{exprs}(n) = \emptyset$$
$$\text{exprs}(\texttt{input()}) = \emptyset$$
$$\text{exprs}(e_1 \ op \ e_2) = \{e_1 \ op \ e_2\} \cup \text{exprs}(e_1) \cup \text{exprs}(e_2)$$

```
// x + 1 is available
x = x + (y + z);
// y + z is available
```

Available Expression Analysis — Constraint Rules (Remaining)

- `if` x: $[\![v]\!] = \text{JOIN}(v)$
- `entry`: $[\![v]\!] = \top = \emptyset$
- `return`: $[\![v]\!] = \text{JOIN}(v)$

Very Busy Expression Analysis

- An expression is *very busy* if it will definitely be evaluated before its value changes

```
// nothing is very busy
x = input();
// x + 1 is very busy
if input() {
  // x + 1 is very busy
  y = x + 1;
} else {
  // x + 1 and y + 1 are very busy
  z = x + 1;
  // y + 1 is very busy
  w = y + 1;
}
```

Very Busy Expression Analysis — Motivation

- We can approximate the set of very busy expressions using dataflow analysis
  - Application: optimization (code hoisting)
  - We want: the answer "very busy" can be trusted and "not very busy" is safe but useless

Very Busy Expression Analysis — Optimization Example (if)

Before:

```
x = input();

if input() {
  y = x + 1;
} else {
  z = x + 1;
  w = y + 1;
}
```

After:

```
x = input();
xplus1 = x + 1;
if input() {
  y = xplus1;
} else {
  z = xplus1;
  w = y + 1;
}
```

Very Busy Expression Analysis — Optimization Example (while)

Before:

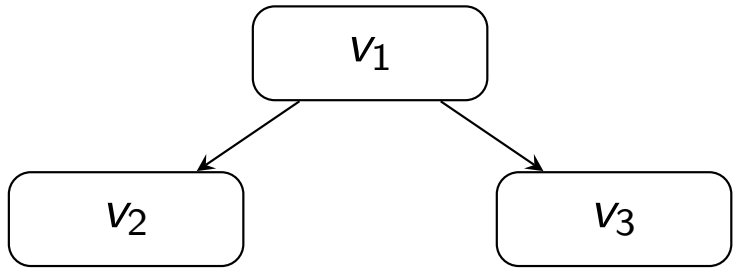
```
x = input();

while input() {
  y = x + 1;
}
z = x + 1;
```

After:

```
x = input();
xplus1 = x + 1;
while input() {
  y = xplus1;
}
z = xplus1;
```

Very Busy Expression Analysis — Abstract States

$[\![v_1]\!] = \{x + 1\}$



$[\![v_2]\!] = \{x + 1, y + 1\}$ $\quad$ $[\![v_3]\!] = \{x + 1\}$

- State $= (\mathcal{P}(\text{Expr}), \supseteq)$ — **Reverse** power set lattice
- For each CFG node v, $[\![v]\!]$ denotes the set of expressions very busy before the node
- $\text{JOIN}(v) = \bigsqcup_{u \in \text{succ}(v)} [\![u]\!] = \bigcap_{u \in \text{succ}(v)} [\![u]\!]$
  - This combines abstract states from the **successors**

Very Busy Expression Analysis — Constraint Rules

- x=e: $[\![v]\!] = (\mathsf{JOIN}(v)\downarrow x) \cup \mathsf{exprs}(e)$

```
// x + (y + z) and y + z are very busy
x = x + (y + z);
// x + 1 is very busy
```

- `if` x: $[\![v]\!] = \mathsf{JOIN}(v)$
- `entry`: $[\![v]\!] = \mathsf{JOIN}(v)$
- `return`: $[\![v]\!] = \top = \emptyset$

Reaching Definition Analysis

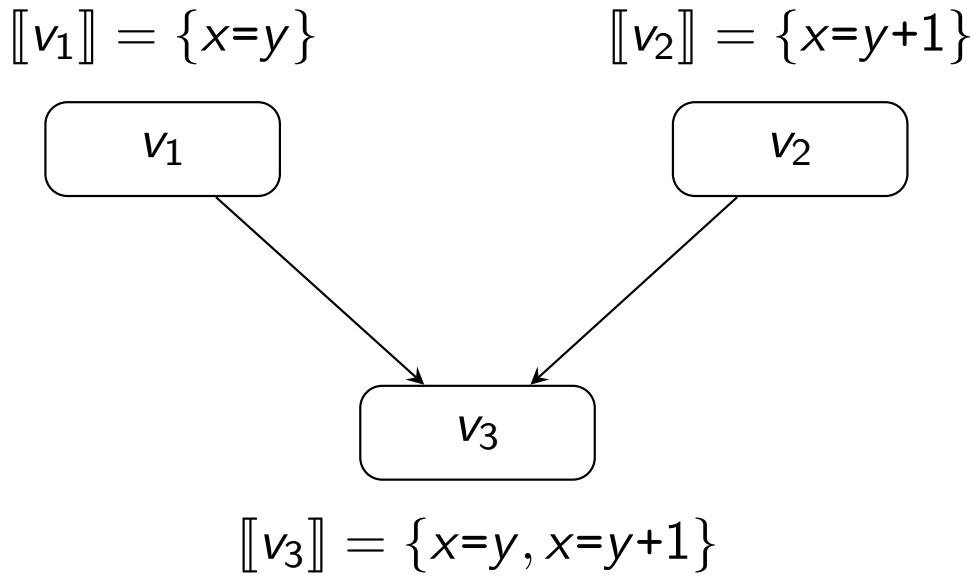
- *Reaching definitions* for a program point are those assignments that may have defined the current values of variables

```
if input() {
  x = y;
  // x = y is a reaching definition
  x = y + 1;
  // x = y + 1 is a reaching definition
} else {
  x = z + 1;
  // x = z + 1 is a reaching definition
}
// x = y + 1 and x = z + 1 are reaching definitions
return x;
```

Reaching Definition Analysis — Motivation

- We can approximate the set of reaching definitions using dataflow analysis
  - Application: def-use graph (useful for optimizations)
  - We want: the answer "not reaching" can be trusted and "reaching" is safe but useless

Reaching Definition Analysis — Abstract States



- State $= (\mathcal{P}(\text{Def}), \subseteq)$ — Power set lattice
  - Def $d ::= x{=}e$
- For each CFG node v, $[\![v]\!]$ denotes the set of definitions that may define values of variables at the program point after the node
- $\text{JOIN}(v) = \bigsqcup_{u \in \text{pred}(v)} [\![u]\!] = \bigcup_{u \in \text{pred}(v)} [\![u]\!]$
  - This combines abstract states from the **predecessors**

Reaching Definition Analysis — Constraint Rules

- x=e: $[\![v]\!] = (\text{JOIN}(v)\downarrow x) \cup \{x\text{=}e\}$
  - $\downarrow x$ removes all definitions of x

```
// x = y + 1 is a reaching definition
x = y;
// x = y is a reaching definition
```

- `if` x: $[\![v]\!] = \text{JOIN}(v)$
- `entry`: $[\![v]\!] = \text{JOIN}(v) = \emptyset$
- `return`: $[\![v]\!] = \text{JOIN}(v)$

Time Complexity

- For SimpleWorkListAlgorithm, if $|\text{dep}(v)|$ is bounded by a constant for all nodes v, the worst-case time complexity is $O(n \cdot h \cdot k)$ where
  - n is the number of CFG nodes
  - h is the height of the lattice $L = \text{State}$
  - k is the worst-case time required to compute f_i

- $O(n \cdot m^2)$ where
  - n is the number of CFG nodes
  - m is the number of variables/expressions/definitions
  - Because $h = m$, $k = O(m)$

Forward vs Backward Analyses

- A *forward* analysis computes information about the past behavior
  - Examples: sign analysis, constant propagation analysis, available expression analysis, reaching definition analysis
  - The analysis starts at the entry node and propagates information forward in the CFG
  - JOIN is defined using pred
  - dep = succ

- A *backward* analysis computes information about the future behavior
  - Examples: live variables analysis and very busy expressions analysis
  - The analysis starts at the return node and propagates information backward in the CFG
  - JOIN is defined using succ
  - dep = pred

May vs Must Analyses

- A *may* analysis describes information that may possibly be true
  - Examples: live variables analysis and reaching definitions analysis
  - Typically uses a power set lattice
- A *must* analysis describes information that must definitely be true
  - Examples: available expression analysis and very busy expression analysis
  - Typically uses a reverse power set lattice

May vs Must Analyses — Soundness

- May $\neq$ Sound, Must $\neq$ Complete
- All these analyses are sound but not complete

May vs Must Analyses — Soundness (Live Variables)

- Live variables $= \{x\}$
  - Set of possible behavior: any execution that does not require any variable other than x to be live
    - Can have false positives (some such executions may be actually impossible)
  - Set of impossible behavior: any execution that requires some variable other than x to be live
    - No false negatives (such executions are indeed impossible)

May vs Must Analyses — Soundness (Available Expressions)

- Available expressions $= \{x + y\}$
  - Set of possible behavior: any execution that has already computed $x + y$
    - Can have false positives (some such executions may be actually impossible)
  - Set of impossible behavior: any execution that has not computed $x + y$
    - No false negatives (such executions are indeed impossible)

Classification of Dataflow Analyses

	Forward	Backward
May	Reaching definition analysis	Live variable analysis
Must	Available expression analysis	Very busy expression analysis

Example — Initialized Variable Analysis

```
if input() {
  x = 1
  // x is initialized
  y = x + 1
  // x and y are initialized
} else {
  y = 2
  // y is initialized
}
// y is initialized
z = y + x
```

Example — Initialized Variable Analysis (cont.)

- We want to know whether a certain variable is definitely initialized at a program point
  - Must analysis — State $= (\mathcal{P}(\text{Var}), \supseteq)$
- Initialization is a property of past
  - Forward analysis — $\text{JOIN}(v) = \bigsqcup_{u \in \text{pred}(v)} [\![u]\!]$

- x=e: $[\![v]\!] = \text{JOIN}(v) \cup \{x\}$
- `if` x: $[\![v]\!] = \text{JOIN}(v)$
- `entry`: $[\![v]\!] = \emptyset$
- `return`: $[\![v]\!] = \text{JOIN}(v)$

Transfer Functions

- All constraint functions are of the form $[\![v]\!] = t_v(\text{JOIN}(v))$ where $t_v : L \to L$
- Example: live variable analysis
  - x=e: $[\![v]\!] = \text{JOIN}(v) \setminus \{x\} \cup \text{vars}(e)$
  - `if` x: $[\![v]\!] = \text{JOIN}(v) \cup \{x\}$
  - `entry`: $[\![v]\!] = \text{JOIN}(v)$
  - `return`: $[\![v]\!] = \text{JOIN}(v)$

Transfer Functions (cont.)

- t_v is called a *transfer function* for the CFG node
  - Forward analysis: input to the transfer function represents the state immediately before the node, and the output represents the state immediately after the node
  - Backward analysis: input to the transfer function represents the state immediately after the node, and the output represents the state immediately before the node
- Example: live variable analysis
  - $t_{x=e}(s) = s \setminus \{x\} \cup \text{vars}(e)$

Transfer Functions — Redundancy in SimpleWorkListAlgorithm

- In SimpleWorkListAlgorithm, $\text{JOIN}(v) = \bigsqcup [\![u]\!]$ is computed in each iteration
- However, $[\![u]\!]$ often has not changed since last iteration, so much of the computation is redundant

- We can use transfer functions to avoid redundancy
- Now, $x_i = [\![v_i]\!]$ is the state *before* v_i in forward analyses, and the state *after* v_i in backward analyses

PropagationWorkListAlgorithm

PropagationWorkListAlgorithm$(t_1, \ldots, t_n)$**:**

- $x \leftarrow \bot$
- $W \leftarrow \{v_1, \ldots, v_n\}$
- **while** $W \neq \emptyset$ **do**
  - $v_i \leftarrow W$.removeOne()
  - $y \leftarrow t_{v_i}(x_i)$
  - **for** $v_j \in dep(v_i)$ **:**
    - $z \leftarrow x_j \sqcup y$
    - **if** $x_j \neq z$ **:**
      - $x_j \leftarrow z$
      - W.add(v_j)
- **return** x

PropagationWorkListAlgorithm — Intuition

- This computes the same result
- Intuition:
  - SimpleWorkListAlgorithm computes $x_3 = t_1(x_1) \sqcup t_2(x_2)$
  - PropagationWorkListAlgorithm computes $x_3 = x_3 \sqcup t_1(x_1)$ and $x_3 = x_3 \sqcup t_2(x_2)$, which is $x_3 = x_3 \sqcup t_1(x_1) \sqcup t_2(x_2)$
  - If f is monotone and $g(x) = f(x) \sqcup x$, then $\mathrm{lfp}(g) = \mathrm{lfp}(f)$

Summary

- Live variable analysis determines which variables may be needed in the future (backward, may analysis)
- Available expression analysis determines which expressions have already been computed (forward, must analysis)
- Very busy expression analysis determines which expressions will definitely be evaluated (backward, must analysis)
- Reaching definition analysis determines which assignments may define current variable values (forward, may analysis)
- Dataflow analyses are classified along two axes: forward/backward and may/must
- PropagationWorkListAlgorithm avoids redundant JOIN recomputation by propagating transfer function results incrementally